Chapter 16

Real and Quaternionic Representations

We have studied actions of finite groups on complex vector spaces. Now we consider the real case and the relationship between real and complex representations.

16.1 Real Representations

Let G be a finite group. A real representation of G is a real vector space V_0 together with a group homomorphism $G \to \mathrm{GL}(V_0)$ (automorphisms as an $\mathbb{R}$-vector space).

The averaging trick works over $\mathbb{R}$: starting from any inner product on V_0 and averaging over G, we obtain a G-invariant (positive-definite, symmetric) inner product. So if $W_0 \subset V_0$ is a G-invariant subspace, then $W_0^\perp$ is also G-invariant, and every real representation of a finite group is completely reducible.

However, Schur's lemma in its strong form fails over $\mathbb{R}$. For instance, $\mathbb{Z}/n$ acts on $\mathbb{R}^2$ by rotation by $2\pi/n$, which is irreducible (for $n \geqslant 3$) but has nontrivial endomorphisms: any rotation commuting with rotation by $2\pi/n$ works, so $\mathrm{End}_G(\mathbb{R}^2) \cong \mathbb{C}$ (not just $\mathbb{R}$).

16.2 Complexification

The main tool for relating real and complex representations is complexification.

Definition 16.2.1: Complexification

Given a real representation V_0 of G, its complexification is the complex vector space $V = V_0 \otimes_{\mathbb{R}\mathbb{C} = V_0 \oplus i V_0}$, with G acting by $g(v + iw) = gv + i\,gw$ for $v, w \in V_0$. This defines a functor $\{\text{real reps of } G\} \to \{\text{complex reps of } G\}$.

Definition 16.2.2: Real complex representation

A complex representation V of G is called real if there exists a real representation V_0 such that $V \cong V_0 \otimes_{\mathbb{R}\mathbb{C}}$.

Proposition 16.2.1 Necessary condition

If $V = V_0 \otimes_{\mathbb{R}\mathbb{C}}$ is real, then χ_V takes values in $\mathbb{R}$. (The trace of a matrix with real entries is real.)

The converse is also true for irreducible representations, as we will show.

16.3 Characterizing Real Representations

Proposition 16.3.1 Antilinear involution criterion

A complex representation V of G is real if and only if there exists a G-equivariant, complex antilinear map $\tau : V \to V$ (i.e. $\tau(\lambda v) = \bar{\lambda}\tau(v)$ for all $\lambda \in \mathbb{C}$) with $\tau^2 = \mathrm{id}$.

Proof: ($\Rightarrow$) If $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$, define $\tau(v + iw) = v - iw$ for $v, w \in V_0$ (complex conjugation with respect to the real form V_0). Then τ is antilinear, $\tau^2 = \mathrm{id}$, and G-equivariant since g preserves V_0.

($\Leftarrow$) Given such a τ, every $v \in V$ decomposes as

$$v = \underbrace{\frac{v + \tau(v)}{2}}_{\in V^{\tau}} + \underbrace{\frac{v - \tau(v)}{2}}_{\in V^{-\tau}}$$

where $V^{\tau} = \ker(\tau - \mathrm{id})$ is the $+1$-eigenspace and $V^{-\tau} = \ker(\tau + \mathrm{id})$ is the -1-eigenspace. Since τ is antilinear, $\tau(iv) = -i\tau(v)$, so $V^{-\tau} = iV^{\tau}$. Thus V^{τ} is an $\mathbb{R}$-subspace (not a $\mathbb{C}$-subspace!) with $V = V^{\tau} \oplus iV^{\tau} = V^{\tau} \otimes_{\mathbb{R}} \mathbb{C}$. Since τ is G-equivariant, $V_0 := V^{\tau}$ is preserved by G, giving a real representation with $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$. $\square$

16.4 The Frobenius–Schur Indicator

For irreducible representations, we can determine whether a representation is real, quaternionic, or neither using the bilinear form it carries.

Proposition 16.4.1 Invariant bilinear form from self-duality

Let V be an irreducible complex representation of G with χ_V real-valued. Then $\chi_V = \overline{\chi_V} = \chi_{V^*}$, so $V \cong V^*$. By Schur's lemma, $\dim \mathrm{Hom}_G(V, V^*) = 1$, so there is a G-invariant bilinear form $B : V \times V \to \mathbb{C}$, unique up to scaling, and nondegenerate.

Since $B \in (V \otimes V)^* \cong \mathrm{Sym}^2 V^* \oplus \bigwedge^2 V^*$, and both the symmetric and skew-symmetric parts are also G-invariant, uniqueness forces exactly one of them to be zero. So B is either symmetric or skew-symmetric. This dichotomy distinguishes real from quaternionic representations.

Proposition 16.4.2 Real iff symmetric form

An irreducible complex representation V of a finite group G is real if and only if V carries a G-invariant nondegenerate symmetric bilinear form $B : V \times V \to \mathbb{C}$.

Proof: ($\Rightarrow$) If $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$, then V_0 carries a G-invariant real inner product B_0 (by the averaging argument). Extend $\mathbb{C}$-bilinearly:

$$B(v_1 + iw_1,\, v_2 + iw_2) = B_0(v_1, v_2) + iB_0(w_1, v_2) + iB_0(v_1, w_2) - B_0(w_1, w_2).$$

This is a nondegenerate, symmetric, G-invariant $\mathbb{C}$-bilinear form on V.

($\Leftarrow$) Given B, the form determines a G-equivariant isomorphism $\varphi : V \xrightarrow{\sim} V^*$. Choose a G-invariant Hermitian inner product H on V (by averaging), which gives a G-equivariant, antilinear isomorphism $V \xrightarrow{\sim} V^*$ (via $v \mapsto H(-, v)$). Composing one with the inverse of the other yields a G-equivariant, antilinear map $\tau : V \to V$, characterized by $H(\tau(v), w) = B(v, w)$.

Now $\tau^2 : V \to V$ is $\mathbb{C}$-linear and G-equivariant, so by Schur's lemma, $\tau^2 = \lambda \cdot \mathrm{id}$ for some $\lambda \in \mathbb{C}$. We show $\lambda \in \mathbb{R}_{>0}$:

$$H(\tau^2(v), v) = B(\tau(v), v) = B(v, \tau(v)) = H(\tau(v), \tau(v)) \geqslant 0$$

where the second equality uses symmetry of B. So $\lambda \geqslant 0$, and $\lambda > 0$ since $\tau \neq 0$. Replacing H by $\lambda^{-1/2}H$ (or equivalently τ by $\lambda^{-1/2}\tau$), we arrange $\tau^2 = \mathrm{id}$. By the antilinear involution criterion, V is real. $\square$

16.5 Quaternionic Representations

In the other case, the invariant bilinear form B is skew-symmetric. The same argument produces a G-equivariant antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$.

Definition 16.5.1: Quaternionic structure

A quaternionic structure on a complex vector space V is a $\mathbb{C}$-antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$. A representation with such a G-equivariant J is called a quaternionic representation.

Such a J makes V into a module over the quaternions $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$, the division algebra (noncommutative ring where every nonzero element has a multiplicative inverse) with $i^2 = j^2 = k^2 = ijk = -1$. Concretely, $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$ with $ji = -ij$, $j^2 = -1$, so an $\mathbb{H}$-module is a $\mathbb{C}$-vector space with an antilinear map J ("right multiplication by j") satisfying $J^2 = -\mathrm{id}$.

Note:-

A quaternionic representation has χ_V real-valued (since $V \cong V^*$) but does not come from a real representation. The three cases for an irreducible V with real character are:

- V admits a G-invariant nondegenerate symmetric bilinear form $\iff$ V is real $\iff$ $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$.
- V admits a G-invariant nondegenerate skew-symmetric bilinear form $\iff$ V is quaternionic $\iff$ V carries a G-equivariant antilinear J with $J^2 = -\mathrm{id}$.

And if χ_V is not real-valued, then $V \not\cong V^*$ and V is neither real nor quaternionic.

16.6 Examples

Example 16.6.1 (Standard representation of S_3)

The 2-dimensional irreducible V of $S_3 \cong D_3$ is real: S_3 acts on $V_0 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : \sum x_i = 0\} \cong \mathbb{R}^2$ by rotations and reflections, and $V = V_0 \otimes_{\mathbb{R}} \mathbb{C}$.
Alternatively: $V^* \cong V$ (since χ_V is real-valued), and $\bigwedge^2 V^* \cong U'$ contains no trivial summand, so the G-invariant bilinear form must be in $\mathrm{Sym}^2 V^*$. Since $\mathrm{Sym}^2 V^* \cong U \oplus V$ contains a trivial summand, an invariant symmetric form exists, confirming V is real.

Example 16.6.2 (The quaternion group Q_8)

The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$ acts faithfully on $\mathbb{C}^2$ by

$$\pm 1 \mapsto \pm I, \quad \pm i \mapsto \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \pm j \mapsto \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \pm k \mapsto \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The character takes values $\chi(\pm 1) = \pm 2$ and $\chi = 0$ on all other elements—all real. However, this representation is not real: it is quaternionic.
To see this cannot come from a real representation: if Q_8 acted faithfully on $\mathbb{R}^2$, the invariant inner product would give $Q_8 \hookrightarrow O(2)$. But $O(2)$ has only 2 elements squaring to $-I$ (rotations by $\pm 90°$), while Q_8 needs 6 elements ($\pm i, \pm j, \pm k$) with square -1. Contradiction.
The quaternionic structure is: view $\mathbb{C}^2 = \mathbb{H}$ via $(z_1, z_2) \leftrightarrow z_1 + jz_2$, and the above matrices correspond to left multiplication by $\pm 1, \pm i, \pm j, \pm k$. The antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$ is right multiplication by j, which commutes with left multiplication.

Exercise 16.6.1 Exercises

(1) Classify each irreducible representation of S_4 as real, quaternionic, or complex-type. (Hint:

all characters of S_4 are real-valued, so each is either real or quaternionic. Use the dimension criterion: a quaternionic representation must have even dimension.)

(2) Show that if V is a quaternionic representation of G (carrying an antilinear J with $J^2 = -\mathrm{id}$), then $\dim_{\mathbb{C}} V$ is even.

(3) Prove the Frobenius–Schur indicator formula: for an irreducible representation V,

$$\frac{1}{|G|}\sum_{g\in G}\chi_V(g^2) = \begin{cases} 1 & \text{if } V \text{ is real,} \\ 0 & \text{if } \chi_V \notin \mathbb{R}, \\ -1 & \text{if } V \text{ is quaternionic.} \end{cases}$$

(Hint: $\frac{1}{|G|}\sum_g \chi_V(g^2) = \dim(\mathrm{Sym}^2 V)^G - \dim(\bigwedge^2 V)^G$, and exactly one of these dimensions is 1 when $V \cong V^*$.)

(4) Using the Frobenius–Schur indicator, verify that the 2-dimensional irreducible of Q_8 is quaternionic.

(5) Show that the representations Y and Z of A_5 (the 3-dimensional irreducibles with golden-ratio character values) are real representations. (Their characters are real, and they have odd dimension.)

Solution

(1) All characters of S_4 are real-valued (check the character table). A quaternionic representation has $J : V \to V$ with $J^2 = -\mathrm{id}$, and J pairs basis vectors into orbits of size 2, so $\dim V$ must be even. The irreducibles of S_4 have dimensions 1, 1, 2, 3, 3. The odd-dimensional ones (U, U', V_4, V_4') are automatically real. For W ($\dim = 2$): compute the Frobenius–Schur indicator $\frac{1}{24}\sum \chi_W(g^2)$. We have $\chi_W = (2, 0, 2, -1, 0)$ with class sizes $(1, 6, 3, 8, 6)$. Squaring: $e \to e, (12) \to e, (12)(34) \to e, (123) \to (132), (1234) \to (13)(24)$. So $\chi_W(g^2) = (2, 2, 2, -1, 2)$. The indicator is $\frac{1}{24}(2 + 12 + 6 - 8 + 12) = 1$. So W is also real.
(2) The antilinear map J gives an $\mathbb{R}$-linear map $V \to V$ whose square is $-\mathrm{id}$. Viewing V as $\mathbb{R}^{2n}$ (where $\dim_{\mathbb{C}V=n}$), this gives a real matrix M with $M^2 = -I$. The characteristic polynomial $\det(M - tI)$ has degree $2n$ and all roots are $\pm i$, which come in conjugate pairs. More directly: J makes V into an $\mathbb{H}$-module, and $\dim_{\mathbb{H}} V = \frac{1}{2}\dim_{\mathbb{C}} V$ must be a positive integer.
(3) We have $\frac{1}{|G|}\sum_g \chi_V(g^2) = \frac{1}{|G|}\sum_g \mathrm{Tr}(\rho(g^2))$. Using the decomposition $V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V$ and the identities $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$, $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$, we get the indicator $= \dim(\mathrm{Sym}^2 V)^G - \dim(\bigwedge^2 V)^G$. When $V \cong V^*$, the unique invariant bilinear form is either symmetric (contributing 1 to Sym^2) or skew (contributing 1 to $\bigwedge^2$). When $V \not\cong V^*$, neither $\mathrm{Sym}^2 V$ nor $\bigwedge^2 V$ contains a trivial summand, giving 0.
(4) Q_8 has 5 conjugacy classes: $\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}$ of sizes 1, 1, 2, 2, 2. For the 2-dimensional rep: $\chi = (2, -2, 0, 0, 0)$. Squaring: $1^2 = 1$, $(-1)^2 = 1$, $(\pm i)^2 = -1$, $(\pm j)^2 = -1$, $(\pm k)^2 = -1$. So $\chi(g^2) = (2, 2, -2, -2, -2)$. Indicator: $\frac{1}{8}(2 + 2 - 4 - 4 - 4) = -1$. Quaternionic.
(5) Both Y and Z have real characters and odd dimension 3. Since a quaternionic representation must have even complex dimension (exercise 2), odd-dimensional representations with real characters are automatically real.